

Zi Xuan Lim (she/her)

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CAREER OBJECTIVE

Results-driven AI Engineer specializing in **Computer Vision** and **Vision-Language Models (VLMs)**, with a proven track record of optimizing architectures for high-speed real-time deployment. Adept at engineering end-to-end MLOps pipelines, managing Docker containerization, and building scalable backends with FastAPI. Seeking a remote AI Engineering internship to leverage hands-on experience in model quantization and edge optimization to build and ship highly efficient, production-ready AI solutions.

TECHNICAL SKILLS

- **Deep Learning Frameworks:** PyTorch, TensorFlow, Scikit-learn, YOLO, Faster R-CNN, DETR, MobileNet
- **Computer Vision Tools:** OpenCV, MediaPipe, Roboflow, LabelMe, Image Augmentation, Model Evaluation
- **Multimodal & Generative AI:** Vision-Language Models (VLMs), Qwen-VL, Qwen-Image-Edit
- **Deployment & DevOps:** FastAPI, Gradio, Streamlit, Quantization, Docker, Conda, Git/GitHub, Linux/Ubuntu
- **Programming & Databases:** Python, Java, Scala, SQL (Oracle/MySQL), MongoDB
- **Hardware:** UAV Assembly, Circuit Integration

PROFESSIONAL EXPERIENCE

WISE AI Malaysia Sdn. Bhd. | Bandar Sunway, Selangor

AI Engineer Intern

January 2026 – April 2026

- **Face Quality & Eye-Closure Classification:** Developed and fine-tuned an advanced face quality assessment model, engineering a robust eye-closing classifier utilizing MobileNet and MediaPipe. Spearheaded the systematic evaluation of these models alongside Qwen-VL and YOLO architectures
- **Interactive Deployment & Visualization:** Deployed the eye-state classification model using Gradio, creating an interactive, real-time visualization interface to effectively monitor and validate user eye-closure behaviors.
- **AutoML & MLOps Infrastructure:** Designed a FastAPI-based backend for automated data ingestion and model preprocessing workflows specifically tailored for facial image datasets. Managed environment containerization and GPU resource allocation within Linux ecosystems using Docker.
- **Edge Optimization & High-Speed Deployment:** Implemented Post-Training Quantization (PTQ) using FP16 and INT8 formats to optimize heavy face quality models for deployment via ONNX and OpenVINO, achieving a 40% gain in inference speed.
- **Team Mentorship & eKYC Pipelines:** Supervised and mentored an incoming intern, providing comprehensive training and direction on complex data annotation pipelines for ID fraud detection, face liveness, and eKYC systems.

Research Centre for Human-Machine Collaboration (HUMAC) | Sunway University

Lab Assistant

June 2024 – January 2026

- **Academic Support:** Assisted PhD candidate in research project integrating computer vision into UAVs for Malaysian flood rescue operations
- **UAV Integration:** Designed and assembled a custom-built drone, including hardware setup, circuit integration, and flight software configuration to support data collection for research
- **Algorithm Benchmarking:** Conducted comparative research on object detection architectures, including YOLO, Faster R-CNN, and Detection Transformers (DETR), to determine optimal accuracy-latency trade-offs for real-time UAV applications
- **Research Leadership & Publications:** Co-led brainstorming, ideation, and task delegation for smart city traffic management research, resulting in a co-first authored publication in a **Q2 journal** (Journal of Electrical and Computer Engineering). Currently spearheading a second systematic review on flood response technologies targeting a **Q1 journal** (IEEE Access).

NOTABLE PROJECTS

Standalone and Open-Source Vehicle Detection Framework Using YOLO for Traffic Analysis

Bachelor's Degree Final Year / Capstone Project

April 2025 – January 2026

- Addressed critical gaps in Southeast Asian ITS caused by the lack of localized datasets and the failure of Western-centric models to generalize to local vehicle brands (e.g., Proton, Perodua) and tropical weather conditions

- Fine tuned and identified YOLO11s as the optimal architecture, achieving a mean Average Precision (mAP) of 0.774, an F1-score of 0.742, and a high-performance real-time inference speed of **666 FPS**.
- Conducted in direct collaboration with the **Malaysian Public Works Department (JKR)** to establish a robust, open-source benchmark for future privacy-compliant ITS research in the region

MY-VID: An Open-Source Malaysian Vehicle Image Dataset

Research Project

December 2024 – December 2025

- Developed the first open-source dataset of Malaysian road vehicles, addressing the lack of localized datasets for intelligent transportation academic research
- 9,000 annotated images of Malaysian vehicles under diverse conditions. Applied dataset in traffic management and road safety applications, including vehicle detection, congestion prediction, and accident analysis in Malaysia

ACADEMIC ACHIEVEMENTS

Sunway University

Bachelor of Science (Honours) in Computer Science

Bandar Sunway, Selangor

April 2023 - April 2026

CGPA: 3.98/4.00

Relevant Coursework: Machine Learning, Computer Vision, Data Mining, Predictive Modeling, Model Evaluation

Honor: Jeffrey Cheah Foundation (JCF) Scholarship (**100% Tuition Fee Waiver** for outstanding academic performance)

Sunway College Kuala Lumpur

Foundation in Science and Technology (FIST)

Bandar Sunway, Selangor

April 2022 - March 2023

CGPA: 4.00/4.00

Specialization: Completed the **Computer Science track**, focusing on algorithmic foundations and technical sciences

Honor: Awarded **Sunway University President's Award** to the highest scoring student

Sekolah Menengah Kebangsaan Yong Peng

Sijil Pelajaran Malaysia (SPM)

Batu Pahat, Johor

January 2017 - March 2022

Grades: 10As (5A+, 2A, 3A-)

Technical Science Stream: Specialized in **Computer Science** and **Advanced Mathematics**

Honor: Top Achiever Award for high academic standing among SPM 2021 cohort

PUBLICATIONS & RESEARCH PRESENTATIONS

Journal Manuscripts

- Teoh, J. **Lim, Z. X. (Co-first authors)**, Chan, Y. H., Kabir, A. S., Yap, K. M., "Recent Advancements, Challenges, and Future Directions of Integrating Drones in Smart City Traffic Management: A Systematic Review, *Journal of Electrical and Computer Engineering*, 2026, 7869329, 26 pages, 2026. <https://doi.org/10.1155/jece/7869329>.
- Teoh, J., **Lim, Z. X.**, Zulkoffli, Z. B., Chua, H. S., Lee, Y. L., and Yap, K. M., "A Systematic Review on Flood Response in Malaysia: Technical Advancements, Challenges and Opportunities." targeting *IEEE Access*. (**In Preparation**)

Conference Proceedings & Presentations

- **Lim, Z. X.**, Teoh, J. and Yap K. M., "Advancing Smart Traffic Infrastructure with AI-based Vehicle Detection Across Computing Configurations." *Women Research Engineers Network (WREN) Symposium 2025*, University of Wollongong (UoW), Australia. (**Best Presentation Award**)
- **Lim, Z. X.**, Teoh, J. and Yap K. M., "MY-VID: An Open-Source Malaysian Vehicle Image Dataset for Intelligent Transportation System and Road Safety." *International Workshop on Advanced Image Technology (IWAIT) 2026*. (**Extended Abstract Accepted**)

Open-Source Datasets

- **Lim, Z. X.**, Teoh, J. and Yap K. M., "MY-VID: An Open-Source Malaysian Vehicle Image Dataset for Intelligent Transportation System and Road Safety." *Zenodo* (2025). <https://doi.org/10.5281/zenodo.16866508>.

LEADERSHIP & ACTIVITIES

Sunway Tech Club, Sunway University

Education Department Committee

May 2024 – April 2025

- Collaborated with committee members to brainstorm and propose workshop and event ideas aligned with current trends and interests in technology
- Conducted research and evaluated potential topics, speakers, and activities to enhance members' knowledge in technology domains